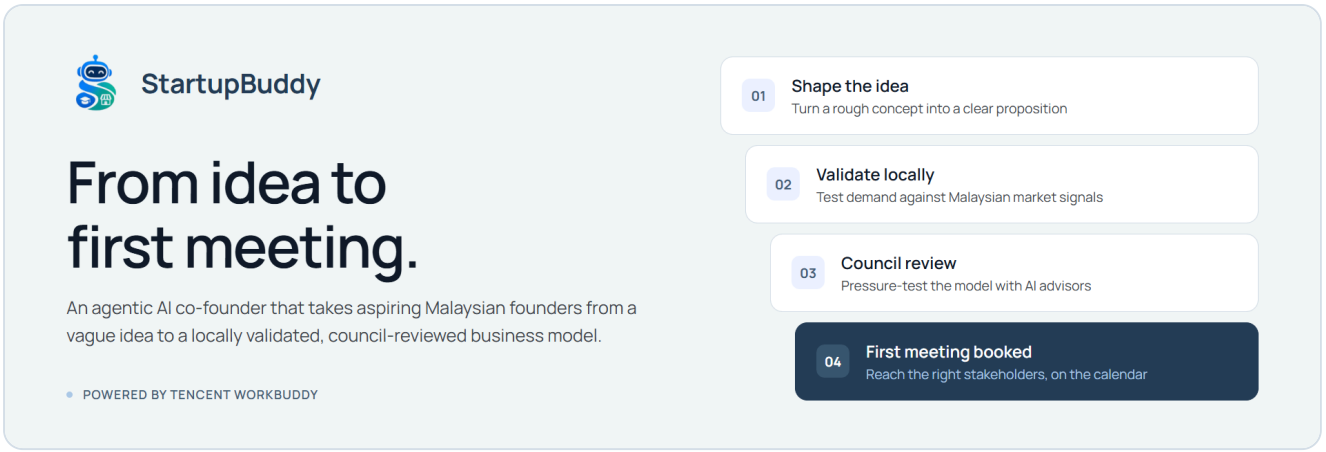


StartupBuddy

From idea to first meeting.

An agentic AI solution that takes aspiring Malaysian founders from a vague business idea to a locally validated, council-reviewed business model and their first booked stakeholder meetings, powered by Tencent WorkBuddy.

Document	Project documentation: Master Sheet, technical documentation, agent roster, MCP contract
Event	Dormathon 2026 · Track 2: WorkBuddy Agentic Solutions
Team	SMILING FACE WITH SUNGLASSES: Nathan Yap (leader), Abraham Pang, Malcolm Sim, Zoe Tan
Live app	https://dormathon-workspace-1.web.app
Code	https://github.com/nathanyap17/startupbuddy
Status date	11 October 2026

Contents

1. Part A · Master Sheet
2. Part B · Technical documentation
3. Appendix 1 · Agent roster
4. Appendix 2 · MCP contract (WorkBuddy-facing)

Evidence labels used throughout: **[n]** published source · **(Built)** verified in the working system · **(Design)** design choice · **(Pending)** not yet verified · **Roadmap** planned, not built.

Part A · Master Sheet

Purpose. The single, authoritative answer sheet for the Dormathon 2026 Track 2 judging criteria (Problem relevance, Agentic capability, WorkBuddy integration depth, Output reliability, Live execution), plus feasibility and the business model. Every figure is sourced or traced to the code or a recorded run. Each section opens with a plain-language summary; what is proven, assumed or not yet verified is stated explicitly.

Project: StartupBuddy - An agentic AI solution that takes aspiring Malaysian founders from a vague business idea to a locally validated, council-reviewed business model and their first booked stakeholder meetings, powered by Tencent WorkBuddy. **Tagline:** From idea to first meeting. **Team:** SMILING FACE WITH SUNGLASSES: Nathan Yap (leader), Abraham Pang, Malcolm Sim, Zoe Tan
Event: Dormathon 2026, Track 2: WorkBuddy Agentic Solutions, 10–11 October 2026 (3-min pitch + 2-min Q&A) **Try it:** https://dormathon-workspace-1.web.app · API: https://startupbuddy-api-960127792850.asia-southeast1.run.app · Code: https://github.com/nathanyap17/startupbuddy
Status date: 2026-10-11 (live on Google Cloud Run + Firebase Hosting; latest build with intake guardrails and follow-ups deployed; repository in sync with GitHub, 19 commits)

How to read the evidence labels

Label	Meaning
[n]	Published source; see the reference list at the end
(Built)	Verified in the working system: code, a recorded run, or the live deployment
(Design)	A design choice or engineering assumption, not an empirical finding
(Pending)	Stated in good faith but not yet verified; listed in section 8
Roadmap	Planned, not built. Never presented as working

Executive summary

In plain words: Many young Malaysians want to start a business but stall at a vague idea: few graduates become entrepreneurs, young people say they don't know where to get help, and formal finance rarely reaches them. StartupBuddy takes one sentence, for example "I want to sell kolo mee near Padungan, Kuching", typed into Tencent WorkBuddy or the web app. It checks the idea against Malaysian law, grounds it in the founder's real neighbourhood, has 16 AI agents build and challenge a full business model, and, only after the founder approves, puts the first site visits in their calendar and drafts their first outreach emails. WorkBuddy does real work in the loop: it starts the run, researches the market, writes the report and relays the founder's decisions. It is live today; the next step is a pilot with one campus cohort.

Criterion (Track 2 rubric)	One-line answer
1. Problem relevance & real-world utility	Aspiring founders lack locally grounded validation and a path to the right people; one session replaces weeks of scattered searching (≈ 4–5 min per full run in our tests) (Built)

Criterion (Track 2 rubric)	One-line answer
2. Autonomous workflow & agentic capability	Intake screener → coordinator → 9 canvas agents in 3 dependent waves → 5-seat council in parallel → advisor; retries, fallbacks and three human checkpoints (Built)
3. WorkBuddy integration depth	WorkBuddy connects as a custom MCP connector, starts the run from one prompt, pulls tasks from a queue (desk research, report), relays G0/G1/G2 decisions and answers questions (Built)
4. Output reliability, validation & verification	Every figure labelled sourced / estimated / needs verification (enforced in code and database); council QC aggregated by code; illegal ideas blocked; DOCX report + real Calendar/Gmail outputs (Built)
5. Live execution, prototype & pitch	One prompt in WorkBuddy → end-to-end run on the deployed app, recorded on 2026-10-10/11 (Built); live demo available at the event

1. Problem relevance and real-world utility

> **In plain words:** Malaysian business is mostly micro-business, and young people rarely make the jump. The tools they find are generic, help is scattered across agencies, and nothing helps them take the first real step.

1.1 Context: who is affected, and where

- MSMEs are 96.2 % of Malaysian establishments (1,292,580), and microenterprises are 77.4 % of MSMEs (2025) [1].
- About 1.4 % of 48,709 public-university graduates were self-employed in the 2010 tracer study [2], and 1.7 % in 2013 against a 5 % government aspiration [3]. (Pending: latest tracer study.)
- Primary user: a final-year student or fresh graduate (e.g. in Kuching) with an idea and no network (Design). *Aisyah*, used in the pitch, is an illustrative persona.

1.2 The burden

Measure	Value
Youths unaware where to seek advice or financial aid	Reported by the 2023 RYSE survey [4] (Pending: primary report)
Youth entrepreneurs financed by financial institutions	6 % (KRI, cited in [5]) (Pending: KRI original)
New SMEs failing within 5 years	"60 %", a commonly cited estimate with weak sourcing [6]; not used in the pitch

1.3 The gaps (why current practice fails)

1. Generic AI validators and plan generators do not use the founder's actual street, competitors or Malaysian licensing [10].
2. MYStartup Single Window aggregates resources and connections [7] but does not analyse an idea or act on it.
3. Nothing turns a plan into a first meeting with the founder's consent.

1.4 Real-world utility (what a session delivers)

Before	With StartupBuddy (recorded runs, 2026-10-10/11)
Guessing the market	8 kolo mee stalls within 1 km of Jalan Padungan found via Google Places (Built)
Generic plans	A 9-block canvas tied to the place, reviewed by 5 seats (Built)
Not knowing which rules apply	Licences named up front, e.g. MBKS food premises licence, food handler training, typhoid vaccination (Built)
No first step	Calendar site visits, Gmail drafts and checklist tasks, created after approval (Built)
Weeks of searching	≈ 4–5 min per full run (202 s and 277 s measured locally with both checkpoints auto-approved) (Built)

2. Autonomous workflow and agentic capability

> **In plain words:** One goal goes in; agents understand it, split the work, run in parallel where they can, check each other and fix what fails. The founder only steps in at checkpoints.

2.1 Pipeline (Built)

Phase	Agents	Tools	Checkpoint
0 · Intake	Intake Screener	Malaysian law knowledge + code backstops	**G0** clarification, only if ambiguous; never auto-approved
1 · Ground	Problem Coordinator	Google Places (physical/hybrid only), knowledge search, WorkBuddy desk research	**G1** confirm or edit the problem
2 · Build	9 BMC agents in waves ① 2 → ② 5 → ③ 2	knowledge search	
3 · Challenge	5 council seats in parallel (Customer, CEO, Investor, CFO, Local Mentor)	deterministic aggregation in code	
3b · Report	Tencent WorkBuddy	report skill; backend renders DOCX	
4 · Act	Advisor	Google Calendar, Meet, Gmail drafts	**G2** approve each action (or opt-in Always allow)
5 · Follow up	Follow-up reader	Gmail read-only, Calendar	consent given at G2

16 agents do the core work (1 coordinator + 9 BMC + 5 council + 1 advisor); helper agents handle intake screening, reply reading, chat and the pilot summary. All run on Gemini 3.5 Flash via Google ADK 2.6.1 with structured (Pydantic) outputs (Built).

2.2 Rubric capabilities

Capability	Evidence
Task decomposition	Intake screener classifies verdict, business type and licences; coordinator decomposes into who / where / pain / demand / competition (Built)

Capability	Evidence
Multi-step planning	5 phases with 3 checkpoints; phase state persisted in Postgres (Built)
Parallel execution	Wave 2 runs 5 agents at once; the council runs 5 seats at once (`asyncio.gather`) (Built)
Multi-agent collaboration	Each wave reads earlier blocks; the council reads the whole canvas; the advisor cites the agent each action came from (Built)
Dynamic error handling	Per-turn timeouts; one automatic retry per canvas block; screener retry then deterministic fallback; WorkBuddy task leases, deadlines and max 2 attempts; visible fallbacks (Built)
Adapts to the idea	Physical vs digital: a digital budgeting app made 0 Google Maps calls and researched online competitors instead (Built, local test 2026-10-11)

3. WorkBuddy integration depth

> **In plain words:** WorkBuddy is where the founder starts and talks, and it does real work (research, the report, relaying decisions), not just chat.

- **Connection:** StartupBuddy's MCP server (streamable HTTP, `X-API-Key`, 10 tools) is added in WorkBuddy's MCP Server Management (`mcp.json`); verified with WorkBuddy 5.7.6 (EXP-1, Built) [8] [9].
- **Task queue:** MCP lets WorkBuddy call us but not the reverse, so cloud agents enqueue tasks and WorkBuddy's `startupbuddy-worker` skill long-polls `await_next_task` (≤ 25 s per call, under WorkBuddy's ~30 s tool limit); EXP-2 passed with 10 polls and 3 tasks (Built).
- **Real work done by WorkBuddy in the recorded run:** started the session from one prompt; `desk_research` returned 11 findings (10 sourced) and 6 regulations; `report_builder` wrote a 15.4k-character report; relayed G1/G2 decisions (Built).
- **Skills:** worker (session start, task loop, G0/G1/G2 relay, `ask_session`, `check_followups`), desk research, report builder (Built).
- **Honest boundary:** the 16 agents run on Google Cloud Run; WorkBuddy is the entry point, researcher, writer and decision relay. Roadmap: WorkBuddy executes the approved Google actions through its own connectors.

4. Output reliability, validation and verification

> **In plain words:** You can check every number, wrong or illegal ideas are caught early, and what you get is complete and usable: a report, calendar entries and email drafts.

Check	What it does
Evidence labels	Every figure is `sourced` (needs a URL), `estimated` (needs a method) or `needs_verification`; unsupported labels are downgraded in code and the database rejects violations (Built)
Intake guardrails	"Sell drugs" → G0 clarification; "ganja and pills" → blocked, citing the Dangerous Drugs Act 1952 and Poisons Act 1952; "registered pharmacist, community pharmacy" → regulated with Type A licence (Poisons Act 1952), APC and premises licence (Built, local tests 2026-10-11)
Code backstops	Risky words force clarification even if the model says "allowed"; clearly illegal words block with the law cited; deterministic fallback when the model stalls (Built)

Check	What it does
Council QC	5 seats score all 9 blocks; plain code computes per-block means and the majority verdict; weak blocks get a "Refinement flag" (Built)
Right action, safely	Email outreach first (Gmail **draft**, never sent), site visits without video links, online meetings only when naturally online, tasks as checklist items; follow-up events go into the founder's own calendar with no attendees (Built)
Deliverables	Problem statement + labelled evidence, 9-block canvas, 5 scorecards, pilot summary, Markdown report + DOCX (44 KB, 8 tables in the recorded run), Calendar events, Gmail drafts (Built)
Audit trail	Every thought summary, tool call, handoff and latency is stored and streamed live to the canvas (Built)

5. Live execution and prototype

> **In plain words:** It's deployed and it runs end to end from one prompt; the demo is a real run, not a mock.

Item	Status
Backend: FastAPI + ADK on Cloud Run (`asia-southeast1`), Cloud SQL Postgres, Secret Manager	Live; `/health` ok + db ok (Built, 2026-10-11)
Frontend: landing (anime.js), Studio chat feed, live canvas, decisions pane, report	Live on Firebase Hosting (Built)
One prompt in WorkBuddy → end-to-end run → Calendar + Gmail	Recorded end to end (Built, 2026-10-10/11)
Demo assets	29 s screen-recording cut of the real run; 30 s animated walkthrough (labelled as such) with voice-over (Built)
Code size	≈ 2.8k lines of backend Python, ≈ 1.4k lines of frontend TypeScript (measured 2026-10-11)

6. Feasibility

> **In plain words:** It already runs on mainstream cloud services at small cost per session; what remains is content (the knowledge base) and a pilot.

6.1 Proven vs not proven

Component	Status
Deployment, WorkBuddy connector, task queue, real end-to-end run	Proven (Built)
Google Calendar events and Gmail drafts after approval	Proven on the demo account (Built)
Intake guardrails, physical vs digital adaptation	Proven in local tests (Built)

Component	Status
Reply tracking (Gmail read-only)	Reply parser proven on sample replies (in-person, online, no-time); live tracking needs the read-only permission on the deployed token (Pending)
Verified Malaysian knowledge base	1 document today (Pending)
Automated test suite	None yet; verification has been end-to-end runs and logged experiments (Pending)

6.2 Risks and mitigations

Risk	Likelihood	Mitigation
WorkBuddy offline during a run	Medium	Task deadlines + labelled fallback; the run completes without it
Model stalls or returns invalid output	Medium	Timeouts, one retry, deterministic fallbacks, structured outputs
Hallucinated figures	Medium	Evidence labels enforced in code; council review; founder checkpoints
Credit cost of WorkBuddy polling	Medium	Worker stops after 20 idle polls (≈ 1.4 credits per poll measured); report DOCX moved server-side
Legal misreading	Low–medium	Intake screener + code backstops; "guidance, not legal advice"

6.3 Roadmap

Phase	Milestones
Next 2 weeks	Grow the verified knowledge base (SSM, DBKU/MBKS, food handler, TEKUN, SME Corp, MYStartup, halal, OpenDOSM); enable live reply tracking; add automated tests
Pilot (one campus cohort)	Measure time to first meeting, % figures sourced, meetings held within 30 days, cost per session
After the pilot	Vector search (hybrid retrieval), council revision loop, WorkBuddy-executed Google actions, multi-user accounts

7. Business model

> **In plain words:** Students use it free; universities, incubators and state programmes pay because their founders arrive prepared.

Segment	Role	Value proposition
Students and first-time founders	User	A grounded plan and first meetings in one session, free
Universities, incubators, state entrepreneurship programmes	Payer (target)	Founders who arrive validated, with a reviewed plan and evidence
Lecturers, mentors, programme officers	Influencer	Less time on first-draft reviews

Pricing method (no price quoted yet): `price per session ≈ (Gemini + cloud + WorkBuddy cost) × (1 + margin)`, sold per cohort licence. One full WorkBuddy run used ≈ 77 credits before the report step was optimised; Gemini and Cloud Run cost per session will be measured in the pilot.

Market: 1,292,580 MSMEs in Malaysia [1]. **Ask:** a pilot with one Sarawak campus cohort.

> **Status:** proposed; no customer or partner is signed. All payers are targets.

8. Limitations and open verification items

#	Item	Current state	Action needed
1	Knowledge base	1 verified document	Add ≥ 10 verified Malaysian/Sarawak sources
2	Live reply tracking	Parser tested; deployed token may lack Gmail read-only	Re-link Google with read-only scope, update `gws-token`, test a real reply
3	Automated tests	None	Add pytest for queue, guardrails, evidence labels
4	Tracer-study figures	2010/2013	Find the latest MOHE tracer study
5	RYSE and KRI figures	Secondary reports	Read the primary reports
6	60 % failure rate	Weak sourcing	Not used in the pitch
7	Competitor features	Vendor pages	Hands-on check
8	Run time "≈ 4–5 min"	Local runs with auto-approval	Measure on the deployed stack during the pilot
9	WorkBuddy depth	Agents run on Cloud Run	Move approved Google actions to WorkBuddy connectors

9. Glossary

Term	Plain meaning
Agent	An AI component with one job and specific tools
Business Model Canvas (BMC)	A one-page, 9-block description of how a business works
Checkpoint (G0/G1/G2)	A point where the founder must answer or approve before the agents continue
Council	Five simulated reviewers (Customer, CEO, Investor, CFO, Local Mentor) who score the canvas
MCP	Model Context Protocol: a standard way for AI apps like WorkBuddy to call external tools
Task queue	A list of jobs the cloud agents leave for WorkBuddy to pick up
Evidence label	Sourced / estimated / needs verification, attached to every figure
WorkBuddy	Tencent's desktop AI agent; mandatory for this track

References

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References checked via search summaries on 2026-10-10 (full text not yet read for 2–6). (Built) claims were checked on 2026-10-11 against the live deployment (`/health`, routes), the repository (`backend/app`, `frontend/src`, 19 commits in sync with GitHub), `docs/BUILD_LOG.md` and the recorded demo runs.

Part B · Technical documentation

B1. Methodology: how the agents work

3.1 Pipeline (one goal, end to end)

Phase	Agents	Tools	Checkpoint
0 · Intake	`intake_screener`	Malaysian law knowledge in the prompt + code backstops	**G0** clarification (only if ambiguous; never auto-approved)
1 · Ground	`problem_coordinator`	Google Places (physical/hybrid only), knowledge-base search, WorkBuddy `desk_research`	**G1** confirm or edit the problem statement
2 · Build	9 BMC agents in waves: ① customer segments, value propositions → ② channels, relationships, activities, resources, partners → ③ revenue streams, cost structure	knowledge-base search	
3 · Challenge	5 council seats in parallel + deterministic aggregation (mean per block; majority verdict)		
3b · Report	Tencent WorkBuddy `report_builder`	backend renders the DOCX	
4 · Act	`advisor`	Google Workspace (Calendar, Meet, Gmail drafts)	**G2** approve each action (or Always allow)
5 · Follow up	`follow_up` (reads replies)	Gmail read-only, Calendar	consent given at G2 ("then schedule when they reply")

16 agents do the core work (1 coordinator + 9 BMC + 5 council + 1 advisor). Helper agents handle intake screening, reply reading, chat and the pilot summary. All run on **Gemini 3.5 Flash via Google ADK 2.6.1**, with structured outputs.

3.2 Guardrails and quality control, enforced in code

Guardrail	How
Understand before planning	Intake screener verdicts `allowed / regulated / ambiguous / prohibited`; risky words ("drugs", "judi", "ah long"…) force a clarification even if the model says "allowed"; clearly illegal words block with the law cited (Dangerous Drugs Act 1952, Arms Act 1960…); deterministic fallback if the model stalls
Every figure labelled	`sourced` (needs a URL), `estimated` (needs a method) or `needs_verification`; unsupported labels are downgraded automatically and the database rejects violations
Nothing outward without approval	G1 and G2 checkpoints; Always allow is opt-in and logged; G0 is never auto-approved

Guardrail	How
Drafts, never sends	Gmail actions create drafts only; follow-up events go into the founder's own calendar with no attendees
Right action for the situation	email outreach first for people who haven't agreed to meet; site visits without video links; online meetings only when naturally online
Bounded and resilient	per-agent timeouts, one automatic retry per canvas block, WorkBuddy task leases and deadlines (max 2 attempts), visible fallbacks
Full audit trail	every thought summary, tool call, handoff and latency is stored and streamed live

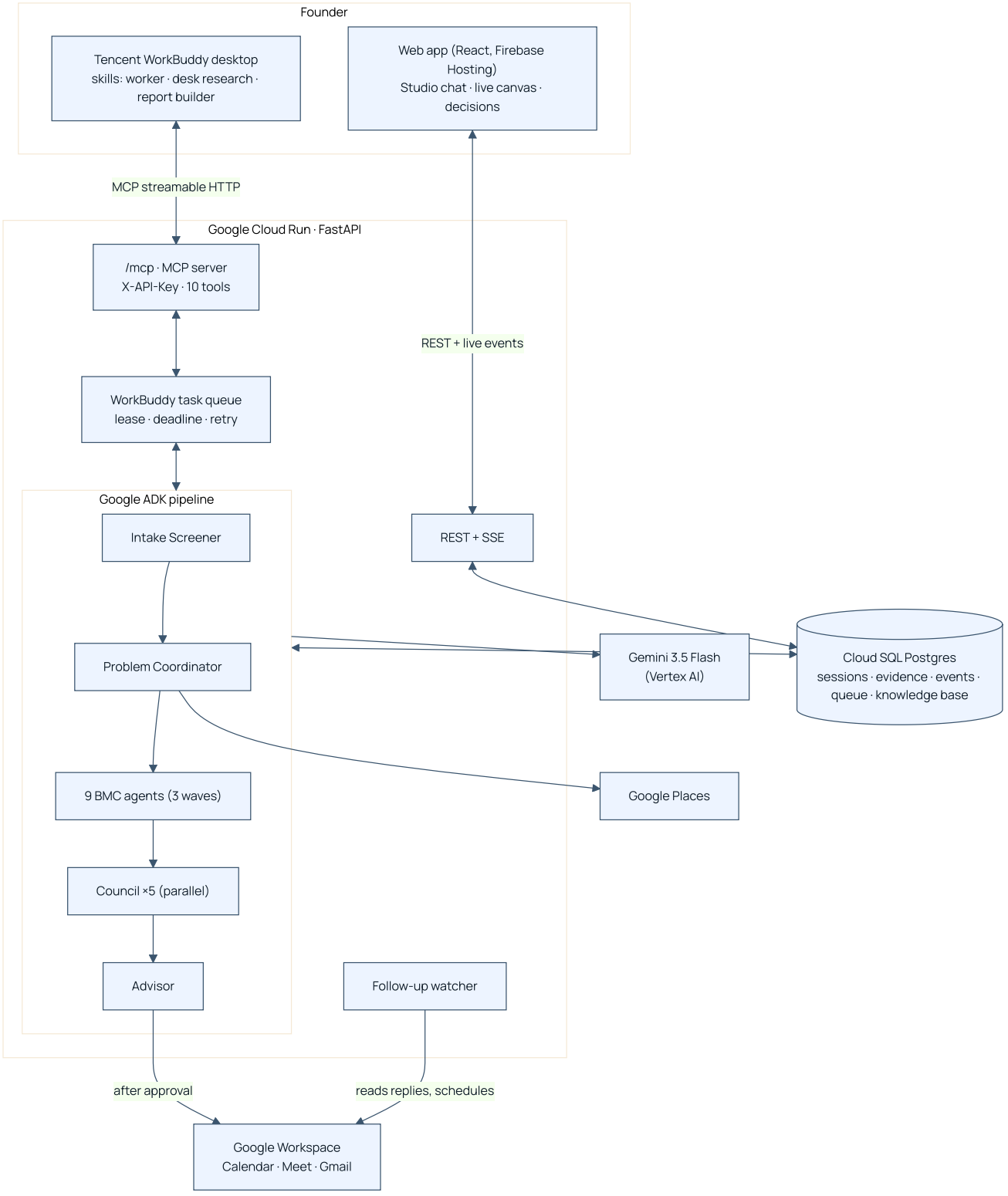
Details: `docs/AGENTS.md`, `.claude/skills/startupbuddy-rules/SKILL.md`.

3.3 Where WorkBuddy fits

WorkBuddy connects to StartupBuddy as a **custom MCP connector**. Because MCP lets WorkBuddy call us but not the other way round, the cloud agents **enqueue tasks** and WorkBuddy's `startupbuddy-worker` skill **long-polls** them (`await_next_task`, ≤ 25 s per call, under WorkBuddy's ~30 s tool limit), does the work with its own skills, and posts structured JSON back (`submit_task_result`, schema-validated). WorkBuddy:

- **starts the run** from one prompt (`start_session`);
- **researches the market** on the web (`desk_research`: findings with sources and labels);
- **writes the full report** (`report_builder`; the backend turns it into a DOCX);
- **relays the founder's decisions** (`confirm_gate` for G0/G1/G2) and answers questions (`ask_session`, `check_followups`).

B2. Architecture

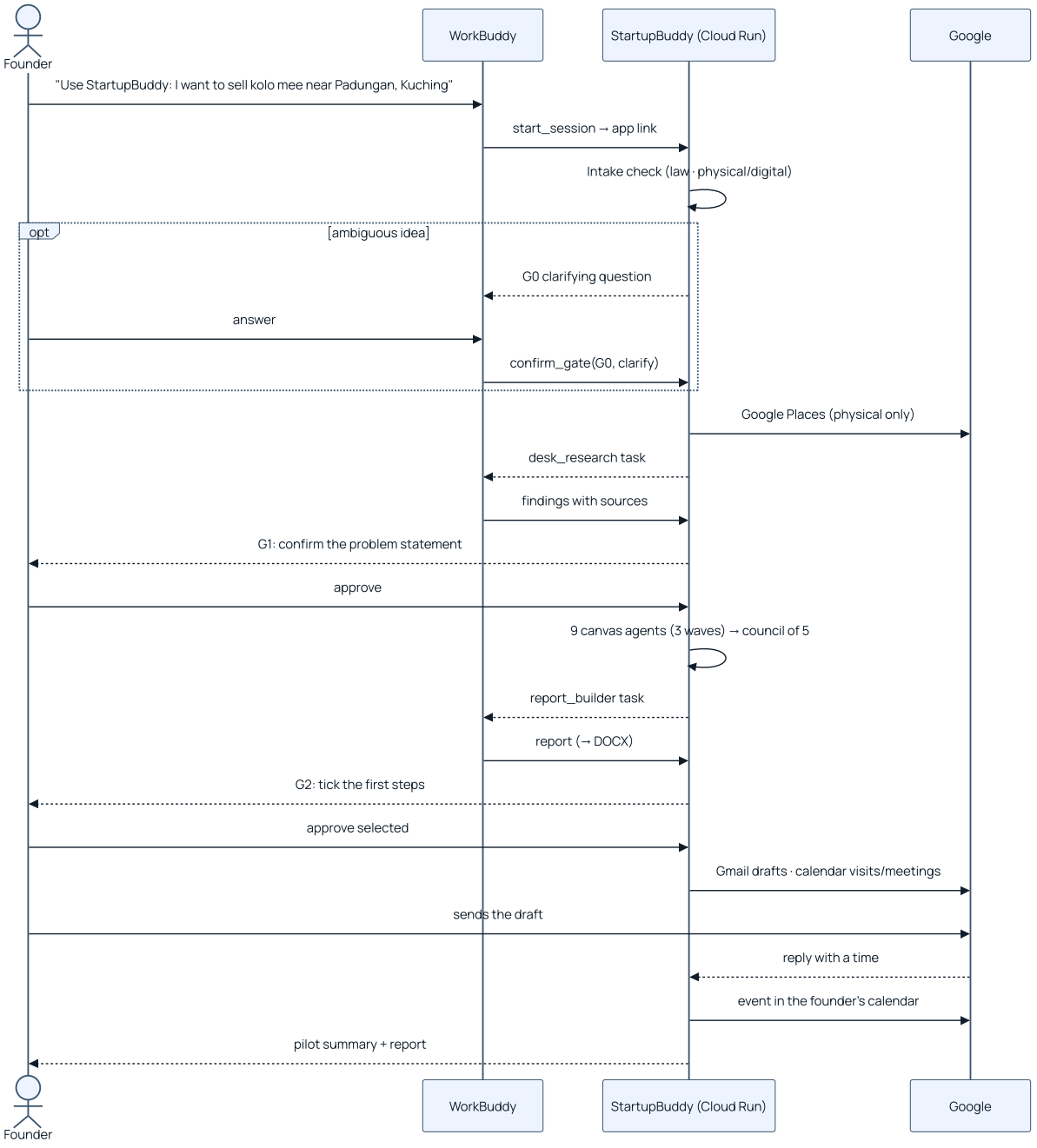


Layer	Technology
Frontend	React 18 + Vite + TypeScript, React Flow (canvas), anime.js (landing), Manrope / "Serene Executive Slate" design (`docs/DESIGN.md`)
Backend	FastAPI, SQLAlchemy 2 async, Pydantic v2, MCP Python SDK 2.2 (streamable HTTP), python-docx
Agents	Google ADK 2.6.1, Gemini 3.5 Flash on Vertex AI (`global`), thinking level LOW with live thought summaries

Layer	Technology
Data	PostgreSQL 16 (Cloud SQL; Docker locally), full-text knowledge search
Integrations	Tencent WorkBuddy (custom MCP connector + skills), Google Places API (New), Google Calendar/Meet, Gmail (drafts + read-only reply tracking)
Hosting	Cloud Run (`asia-southeast1`), Firebase Hosting, Secret Manager

Contracts: `docs/MCP_CONTRACT.md` · data model: `docs/DATA_MODEL.md` · system spec: `docs/PROJECT_SPEC.md`.

B3. User flow



B4. How to use it with Tencent WorkBuddy

Requirements: WorkBuddy desktop (tested on 5.7.6) and the StartupBuddy MCP API key (Secret Manager `mcp-api-key`; ask the team).

1. **Add the MCP server.** In WorkBuddy, open **MCP Server Management** → **Add MCP** and paste the block below. The URL **must end in** `/mcp`. Then **Save**; you should see **startupbuddy** with all tools enabled.

```
{
  "mcpServers": {
    "startupbuddy": {
      "type": "http",
      "url": "https://startupbuddy-api-960127792850.asia-southeast1.run.app/mcp",
      "headers": { "X-API-Key": "<MCP_API_KEY>" }
    }
  }
}
```

2. **Install the skills** (from this repo's root, PowerShell):

```
Copy-Item -Recurse -Force workbuddy\skills\startupbuddy-worker,
workbuddy\skills\startupbuddy-desk-research, workbuddy\skills\startupbuddy-report-builder
"$env:USERPROFILE\.workbuddy-ai\skills\"
```

3. **Start a new WorkBuddy task** (so the connector and skills load) and type one prompt:

> Use StartupBuddy: I want to sell kolo mee near Padungan, Kuching.

4. **Watch and decide.** WorkBuddy shares the app link (open it to watch the canvas), runs its worker loop, and asks you at each checkpoint. You can also ask it things like *"Honestly, is this idea worth it?"* (`ask_session`) or *"Did JAIS reply?"* (`check_followups`).

Good to know: each empty poll costs WorkBuddy credits (≈ 1.4 per poll in our test), so the worker stops after 20 idle polls; a full run used ≈ 77 credits in our first test (before the report step was optimised). Troubleshooting: "SSE error 404" means the URL is missing `/mcp`; "unauthorized" means the key is wrong. More: `docs/WORKBUDDY_INTEGRATION.md`.

B5. How to use the web app

1. Open https://dormathon-workspace-1.web.app → **Open demo app** (single demo account, no login).
2. Type your idea and a Malaysian location → **Start agents**.
3. **Studio (left):** chat with StartupBuddy about your plan; watch each agent's thinking, tool calls and data sources live.
4. **Canvas (centre):** agent cards appear only when their step starts (skeleton → done); tap any card to inspect its content and labelled evidence; the pilot summary appears on top at the end.

5. **Decisions (right):** answer G0 if asked, approve G1 and G2 (or switch on **Always allow**), read the council scorecards, and track your first steps (**Check replies**, **Done**, **Reschedule**, **Retry**). Open the report or download the DOCX.

B6. Impact

Before	With StartupBuddy
Guesswork about the market	Real competitors near the location (e.g. 8 kolo mee stalls within 1 km of Padungan, Google Places) or real online competitors for digital ideas
Generic plans	A 9-block business model tied to the place, reviewed by 5 seats
Five agencies, no starting point	Licences and programmes identified up front by the intake check
Unverifiable numbers	Every figure labelled sourced, estimated or needs verification
No meeting booked	Email drafts, site visits and replies turned into calendar entries
Weeks of scattered searching	≈ 4–5 minutes per full run in our tests (202 s and 277 s, both checkpoints auto-approved)

Pilot KPIs (planned): time from intent to first meeting · % of report figures `sourced` · meetings actually held within 30 days · founder follow-through on `must_verify` items. Aligned with SDG 8 (decent work and economic growth).

B7. Feasibility and what is proven

Component	Status (2026-10-11)
Backend on Cloud Run + Cloud SQL, frontend on Firebase	✅ Live
WorkBuddy MCP connector + long-poll task queue	✅ EXP-1/EXP-2 passed (10 polls, 3 tasks)
Real WorkBuddy end-to-end run (one prompt → research → report → approvals)	✅ Passed: desk research 11 findings (10 sourced)
Google Calendar events and Gmail drafts after approval	✅ Executed on the demo account
Intake guardrails ("sell drugs" → clarify → blocked; pharmacist → regulated)	✅ Tested locally
Digital vs physical (no Maps calls for a digital idea)	✅ Tested locally
Reply tracking (Gmail read-only)	🟠 Parser tested on sample replies; live tracking needs the read-only Gmail permission
Verified Malaysian knowledge base	🟠 Small; being expanded

Evidence log: `docs/BUILD_LOG.md` · experiments: `../sandbox/EXPERIMENTS.md` (workspace).

B8. Commercial model

Users	Students and first-time founders: **free**
Payers (targets)	Universities, incubators and state entrepreneurship programmes, per cohort licence
Why they pay	Founders arrive validated, with a reviewed plan and evidence
Pricing method	price per session ≈ (Gemini + cloud + WorkBuddy cost) × (1 + margin); cost per session to be measured in the pilot
Ask	A pilot with one Sarawak campus cohort

No partner or customer is signed; all payers are targets.

B9. Run locally and deploy

Prerequisites: Python 3.12 + uv, Node 20+, Docker, gcloud CLI (with Application Default Credentials for Vertex AI and Places), Firebase CLI.

```
docker compose up -d db                                  # Postgres on host port 5433
cd backend && cp .env.example .env                       # fill MCP_API_KEY etc. (names in
docs/CREDENTIALS.md)
uv sync && uv run uvicorn app.main:app --port 8000       # API + /mcp
cd ../frontend && npm install && npm run dev             # http://localhost:5173
```

Test without WorkBuddy: the team workspace has `sandbox/sim_worker.py` (not in this repo), a small MCP client that answers WorkBuddy's tasks with canned results.

Deploy (Windows PowerShell, from this folder):

```
powershell -ExecutionPolicy Bypass -File infra\setup-cloud.ps1      # once: DB user, secrets, IAM
powershell -ExecutionPolicy Bypass -File infra\deploy-backend.ps1   # Cloud Run
powershell -ExecutionPolicy Bypass -File infra\deploy-frontend.ps1  # Firebase Hosting
```

Google Workspace (optional): create an OAuth client (Desktop app) as `backend/client_secret.json`, run `uv run python scripts/link_google.py`, then store `backend/token.json` as secret `gws-token` and redeploy. Secrets are never committed; names and locations only in `docs/CREDENTIALS.md`.

B10. Repository structure

```
startupbuddy/
├── backend/
│   ├── app/agent/        # pipeline, prompts, agent factories, assistant, follow-up watcher
│   ├── app/mcp/          # WorkBuddy MCP server (/mcp), Google Workspace actions
│   ├── app/workbuddy/    # task queue, output schemas, DOCX rendering
│   ├── app/routers/      # REST + SSE API
│   ├── app/models/       # SQLAlchemy models (+ idempotent migrations)
│   ├── app/rag/  app/tools/   # knowledge search, Google Places
│   └── scripts/link_google.py
├── frontend/             # React app: landing (/), sessions (/app), workspace (/app/s/:id), report
├── workbuddy/skills/     # WorkBuddy skills: worker, desk research, report builder
├── knowledge/            # verified knowledge-base documents (front matter with source_url)
├── docs/                 # spec, agents, MCP contract, data model, evidence, build log, design
├── infra/                # setup + deploy scripts (PowerShell)
├── Dockerfile  docker-compose.yml  firebase.json
└── CLAUDE.md  PLAN.md  MASTER_SHEET.md
```

B11. Limitations and roadmap

- **Knowledge base:** only a few verified documents so far; next: SSM, DBKU/MBKS licensing, food handler training, TEKUN, SME Corp, MYStartup, halal, OpenDOSM district data.
- **Vector search:** retrieval is full-text today; embeddings (EmbeddingGemma or Vertex) are next for true hybrid search.
- **Council revision loop:** weak blocks are flagged; automatic re-writing (≤ 2 rounds) is designed but not yet switched on.
- **WorkBuddy depth:** WorkBuddy starts runs, researches, writes the report and relays decisions; the 16 agents run on Cloud Run. Next: let WorkBuddy execute approved Google actions through its own connectors.
- **Guidance, not legal advice:** the intake check flags laws and licences; founders should confirm with the regulator.
- **Demo scope:** single demo account, no user registration.

Built for Dormathon 2026 by team SMILING FACE WITH SUNGLASSES. Design: `docs/DESIGN.md` · Brand: `docs/brand/`.

Appendix 1 · Agent roster

AGENTS.md: Agent roster (16 agents)

> Decided 2026-10-10. ADK types refer to `google-adk` (see `.claude/skills/google-adk/SKILL.md`). Every agent can call `search_knowledge` (hybrid RAG). Every tool call is logged to `agent_event`.

Topology

```
StartupPipeline (custom BaseAgent orchestrator; reads gates from DB, so it is resumable)
├── Phase 0  intake_screener (guardrail, added 2026-10-11)
│     verdict allowed | regulated | ambiguous | prohibited (Malaysian law) · business_type physical
| digital | hybrid
│     code backstops: RISK words force clarification; ILLEGAL words (Dangerous Drugs Act 1952, Arms
Act 1960…) block;
│     deterministic fallback if the model can't answer
│     → ⏸ G0 (ambiguous only, never auto-approved, ≤2 rounds) · prohibited → phase "blocked" with
legal alternatives
├── Phase 1  problem_coordinator (LlmAgent, router)
│     tools: places_nearby · opendosm_lookup · search_knowledge ·
enqueue_workbuddy_task(desk_research)
│            save_problem_statement · add_evidence
│     → ⏸ G1: user confirms / edits problem statement
├── Phase 2  bmc_waves (SequentialAgent of 3 ParallelAgents)
│     Wave 1 ‖ customer_segments · value_propositions
│     Wave 2 ‖ channels · customer_relationships · key_activities · key_resources · key_partners
│     Wave 3 ‖ revenue_streams · cost_structure
├── Phase 3  council (ParallelAgent of 5) → aggregate_verdict (deterministic BaseAgent)
│     if any block scores < threshold and round < 2 → re-run only those BMC agents with council
notes
│     → enqueue_workbuddy_task(report_builder, financial_sheet, one_pager)
└── Phase 4  advisor (LlmAgent)
      reads: problem statement, canvas, verdict, open `needs_verification` items
      proposes: action_item[] (meeting, email draft, drive folder, checklist)
      → ⏸ G2: user approves per action → executes approved ones via Google Workspace MCP
(MCPToolset)
```

Roster

#	Agent	Phase	Writes	Key tools	Notes
1	`problem_coordinator`	1	`problem_statement`, `evidence_item`	places_nearby, opendosm_lookup, search_knowledge, desk_research (WorkBuddy)	Reverse-engineers the intent into who / where / pain / demand / competition
2	`customer_segments`	2·W1	`bmc_block`	search_knowledge, read_problem	

#	Agent	Phase	Writes	Key tools	Notes
3	`value_propositions`	2·W1	`bmc_block`	search_knowledge, read_problem, places_nearby	Must state the differentiation against named nearby competitors
4	`channels`	2·W2	`bmc_block`	read_blocks(CS,VP)	
5	`customer_relationships`	2·W2	`bmc_block`	read_blocks(CS,VP)	
6	`key_activities`	2·W2	`bmc_block`	read_blocks(CS,VP), search_knowledge	Licences, permits (SSM, local council, food handler)
7	`key_resources`	2·W2	`bmc_block`	read_blocks(CS,VP)	
8	`key_partners`	2·W2	`bmc_block`	search_knowledge (programmes, suppliers)	Organisations only, never scraped individuals
9	`revenue_streams`	2·W3	`bmc_block`	read_blocks(all)	Prices labelled `estimated` with method
10	`cost_structure`	2·W3	`bmc_block`	read_blocks(all), search_knowledge	Rent, wages, licences: labelled
11	`council_customer`	3	`council_review`	read_canvas	"Would I buy this, at this price, here?"
12	`council_ceo`	3	`council_review`	read_canvas	Strategy, focus, execution risk
13	`council_investor`	3	`council_review`	read_canvas, search_knowledge	Market size, defensibility, fundability (MY programmes)
14	`council_cfo`	3	`council_review`	read_canvas	Unit economics, break-even; recomputes arithmetic in code
15	`council_mentor`	3	`council_review`	read_canvas, search_knowledge	Next 30 days; what to verify first
16	`advisor`	4	`action_item`	Google Workspace MCP (after G2), search_knowledge	Takes tasks from the other agents' decisions

Council review shape (each of 5)

`{ seat, scores: {block_key: 1-5}, strengths[], risks[], must_verify[], improve[], verdict: go|revise|stop }`. Aggregation (code, not LLM): mean per block; a block is sent back if its mean < 3.0 or ≥2 seats flag it in `risks`; overall verdict = majority, ties → `revise`.

WorkBuddy task skills (run by WorkBuddy, not ADK)

Skill	Enqueued by	Output
`desk_research`	problem_coordinator	market notes + source URLs (JSON)
`report_builder`	pipeline after Phase 3	industrial-style report (DOCX/PDF)
`financial_sheet`	pipeline after Phase 3	XLSX: costs, pricing, break-even
`one_pager`	pipeline after Phase 3	PPTX/PDF pitch one-pager (attachment for Advisor emails)

Definitions: `workbuddy/skills/`. Contract: `docs/MCP_CONTRACT.md`.

Guardrail and follow-up helpers (2026-10-11; not counted in the "16 agents")

Agent	When	Output	Notes
`intake_screener`	Phase 0, before anything else	`ScreenOut` (verdict, business_type, needs_location_scouting, licences, critical concerns, clarifying question / refusal)	Shared with every later agent as `screening`; Google Places runs only if `needs_location_scouting`
`follow_up`	after an approved `email_outreach` is sent and a reply arrives	`FollowUpOut` (summary, proposes_time, start, is_online, location, next step)	Watcher checks Gmail every 120 s (needs `gmail.readonly`); a proposed time becomes an event in the founder's own calendar (Meet only if the reply says online; no attendees)
`session_assistant`	founder asks in chat / WorkBuddy `ask_session`	text	Grounded in the session
`pilot_summary`	end of run	one-liner + highlights	

Advisor action kinds: `email_outreach` (draft → wait for reply → schedule), `site_visit` (calendar, no video link → waiting for founder's update), `online_meeting` (calendar + Meet, only when naturally online), `task` (checklist).

Appendix 2 · MCP contract (WorkBuddy-facing)

MCP_CONTRACT.md: WorkBuddy-facing MCP server

> Endpoint: `POST/GET {PUBLIC_BASE_URL}/mcp`, transport **streamable HTTP**. Auth: header `X-API-Key: <MCP_API_KEY>` (WorkBuddy custom connector "API Key" option). Decided 2026-10-10. Constraint: WorkBuddy expects a tool response within ~30 s and a connect timeout of 30 000 ms [WorkBuddy connector docs]. **No tool may block longer than 25 s.**

Why a queue

MCP is client→server only: WorkBuddy calls us; we cannot call WorkBuddy. ADK agents therefore enqueue `workbuddy_task` rows, and WorkBuddy pulls them.

```
agent ──enqueue──► workbuddy_task(status=queued, deadline_at)
WorkBuddy ──await_next_task(wait_s≤25)──► claims oldest queued (FOR UPDATE SKIP LOCKED), lease 120 s
WorkBuddy runs skill  ──heartbeat_task (optional, extends lease)
WorkBuddy ──submit_task_result──► status=succeeded, result JSON validated → agent wakes (asyncio Event / LISTEN-NOTIFY)
lease expires → re-queued (attempts+1, max 2) · deadline passes → status=expired → agent fallback (labelled)
```

Tools

Tool	Args	Returns	Notes
`start_session`	`intent: str`, `location: str`	`{session_id, app_url}`	Entry point for "one prompt" from WorkBuddy. Kicks off Phase 1
`get_session_status`	`session_id`	`{phase, gate, progress, app_url}`	For WorkBuddy to tell the user where things are
`confirm_gate`	`session_id`, `gate: G0|G1|G2`, `decision: approve|edit|reject|clarify`, `notes?`, `action_ids?: list[str]`, `always?`	`{ok}`	Same effect as the web buttons; G0 needs the user's answer in `notes`; G2 requires explicit action ids

Tool	Args	Returns	Notes
`await_next_task`	`worker_id: str`, `wait_s: int = 20` (clamped ≤25)	`{task: null}` or `{task_id, skill, session_id, inputs, output_schema, deadline_at}`	Long-poll. Worker loops until told `stop`
`heartbeat_task`	`task_id`	`{lease_until}`	Extends lease 120 s
`submit_task_result`	`task_id`, `result_json: str`, `artifact_urls?: list[str]`	`{ok, validation_errors[]}`	Rejected if JSON fails `output_schema` → worker may retry
`fail_task`	`task_id`, `reason`	`{ok}`	Agent falls back immediately
`upload_artifact`	`task_id`, `filename`, `content_base64`	`{url}`	Files (DOCX/XLSX/PPTX) stored in GCS bucket; size cap 10 MB
`check_followups`	`session_id`	`{results[]}`	Checks replies to sent outreach emails; replies with a time are added to the founder's calendar (no invites)
`ask_session`	`session_id`, `question`	`{reply}`	Friendly, grounded Q&A about the session (Session Assistant); ≤24 s
`get_session_summary`	`session_id`	markdown	Lets WorkBuddy narrate results in chat

Tool parameters are **flat scalars/lists** (ADK/MCP pattern; avoid a lone model parameter).

Task skill output schemas

Defined as Pydantic models in `backend/app/workbuddy/schemas.py` and exported as JSON Schema in `output_schema`. Every factual item has `{claim, value, label: sourced|estimated|needs_verification, source_url?, method?}`.

Google Workspace MCP (internal, Advisor only)

Mounted at `/mcp-gws`, **not** added to WorkBuddy. Tools: `create_calendar_event(with_meet)`, `create_gmail_draft`, `create_drive_folder`, `upload_to_drive`. Each call must carry an `action_id` whose `approval.status = approved`; the server checks this in code and refuses otherwise. OAuth: demo Google account, consent screen in Testing mode, refresh token in Secret Manager.